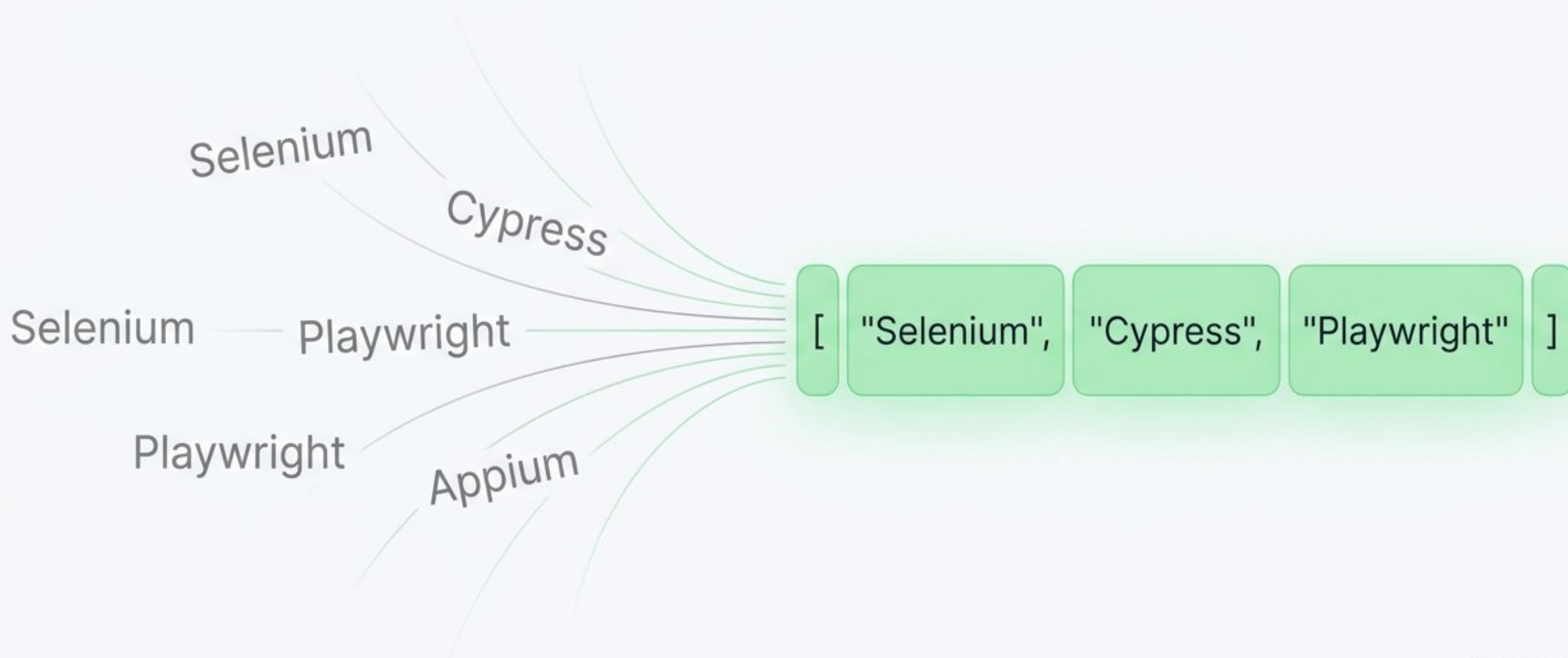


Mastering JavaScript Arrays

The core data structure for modern test automation.



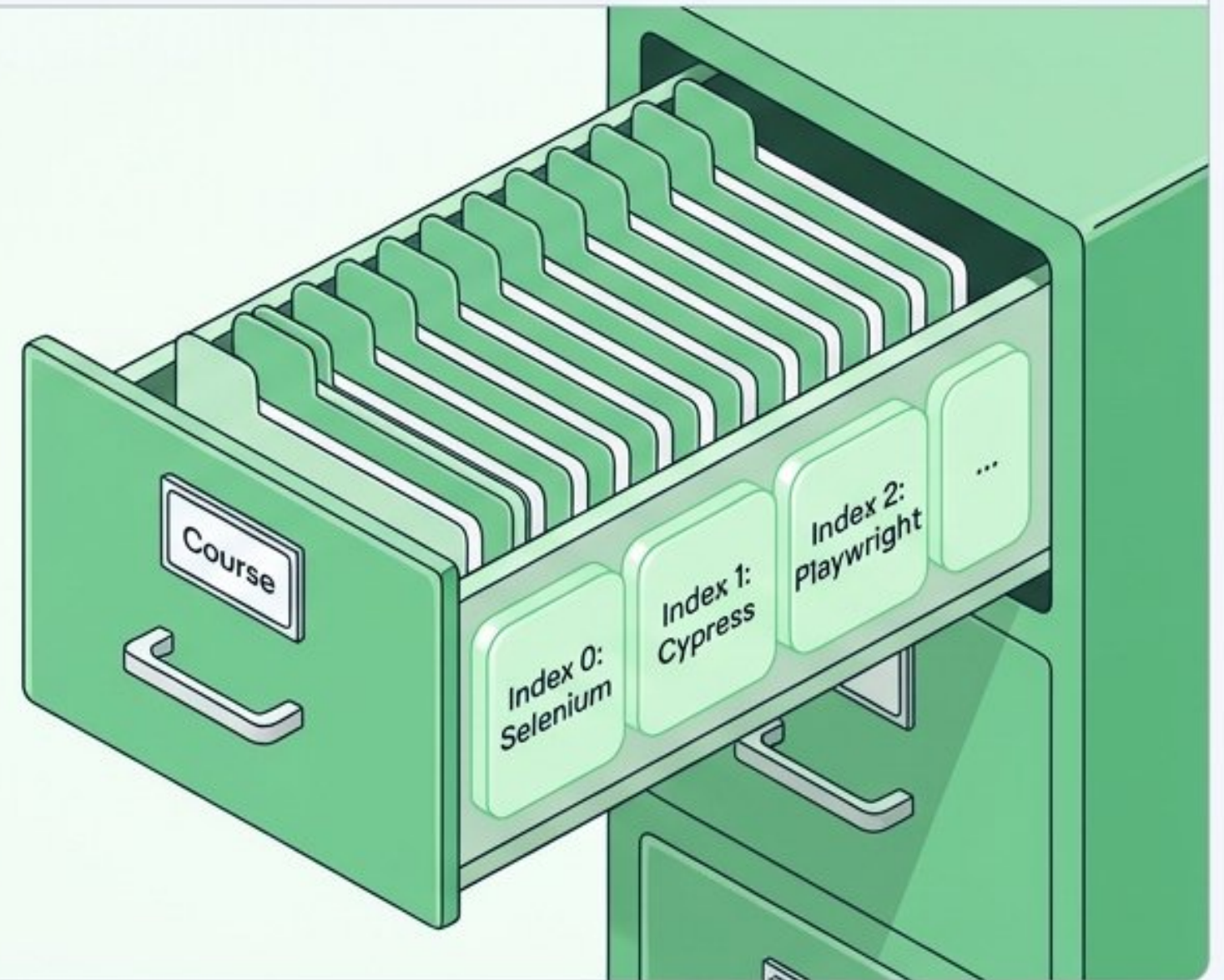
The Variable Trap vs. The Array Solution

Avoid variable clutter. Group related data into a single collection to simplify looping and manipulation.

The Variable Trap



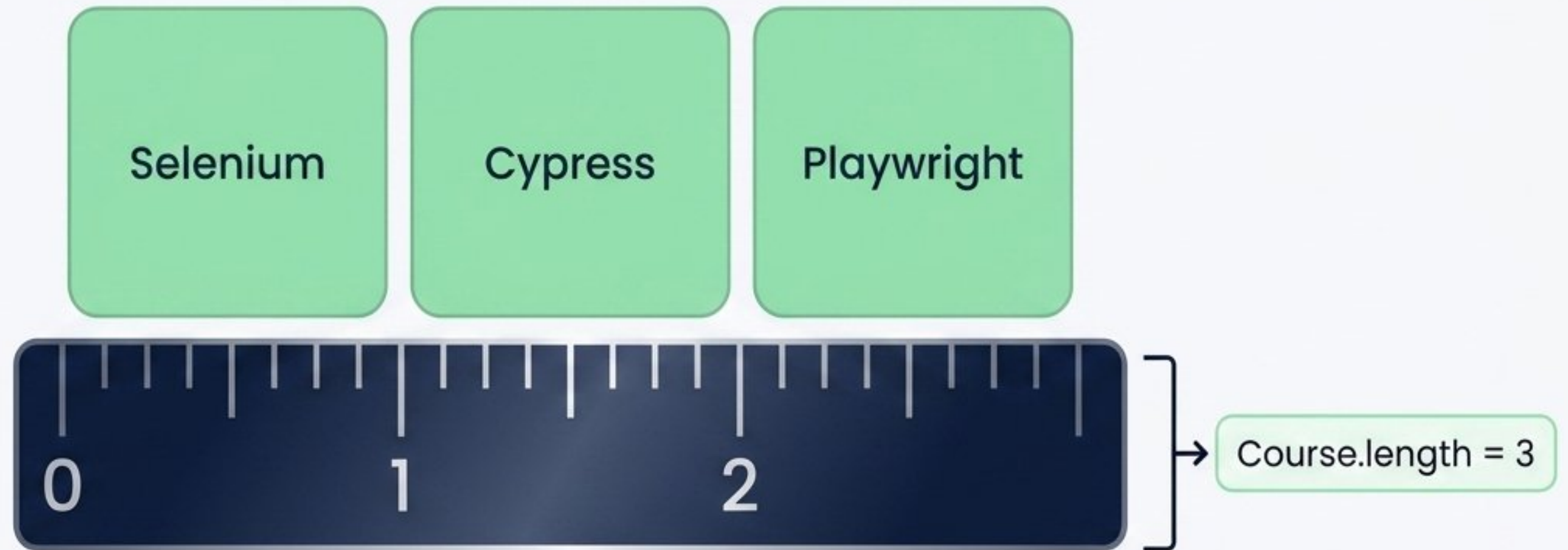
The Array Solution



The Anatomy of an Array

Values are stored in index positions. The count always begins at 0.

```
let Course = ["Selenium", "Cypress", "Playwright"];
```

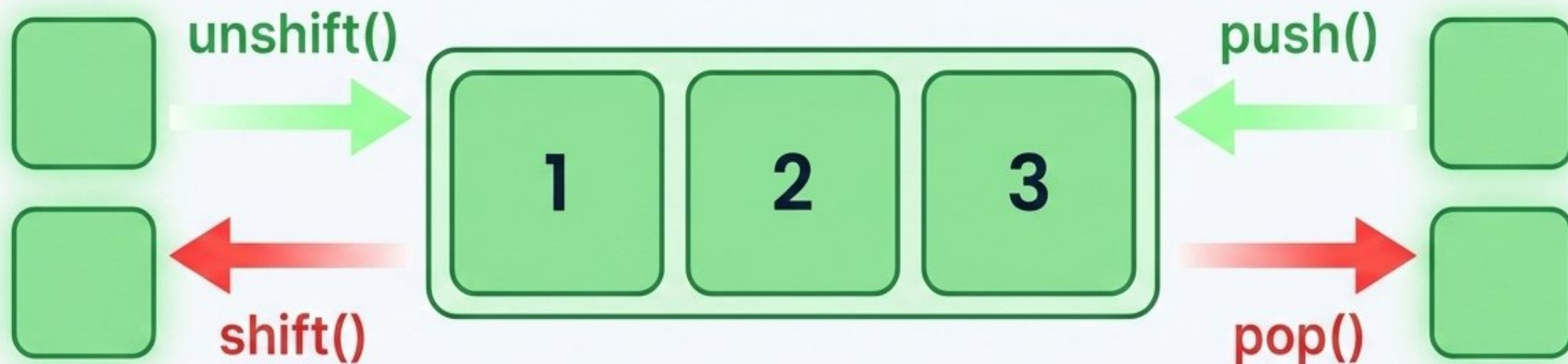


```
Course[0] // "Selenium"
```

```
Course[1] // "Cypress"
```

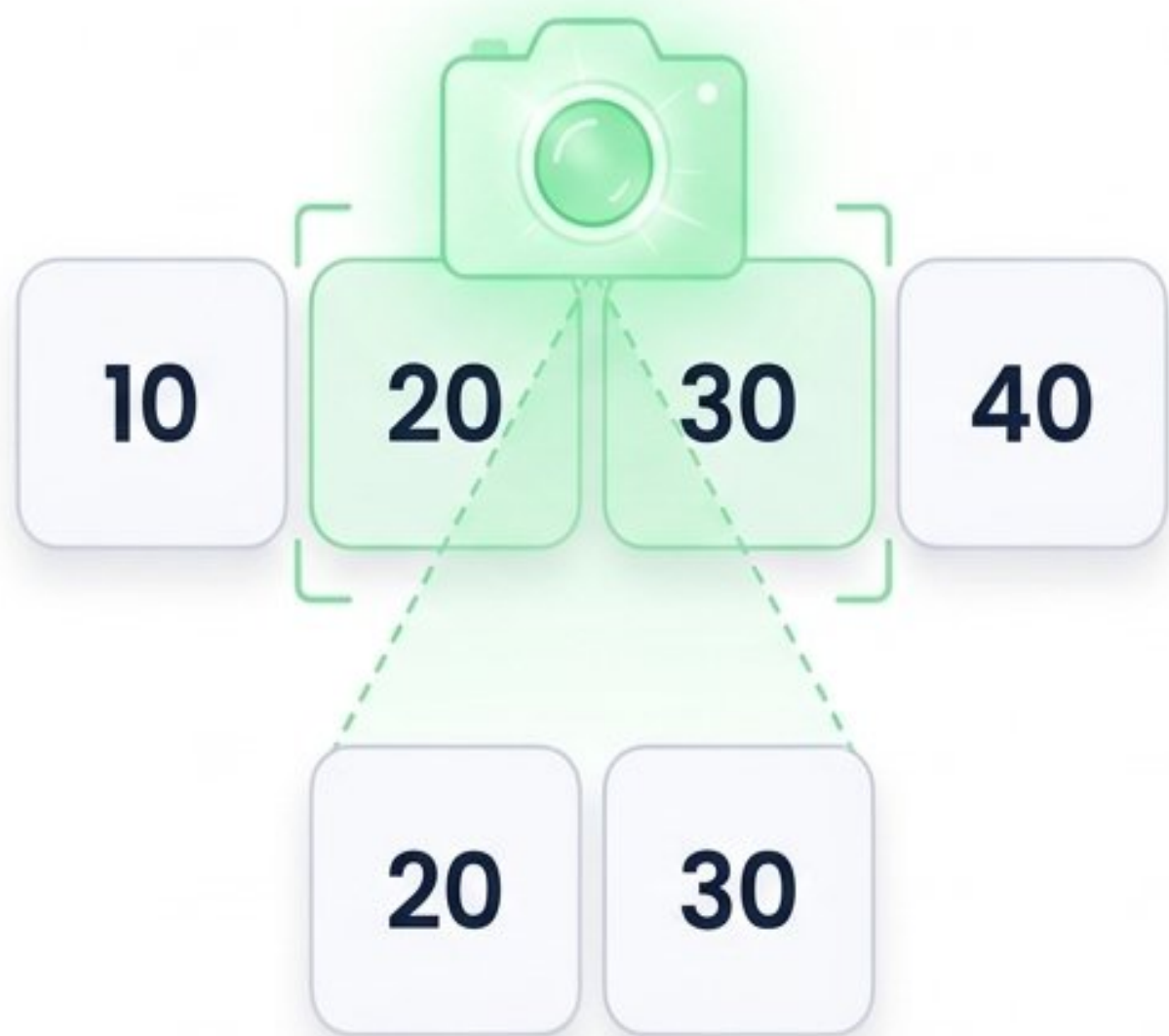
Data Traffic Control: Managing the Boundaries

Easily add or remove elements from the exact beginning or end of your collection.



Modifying the Middle

slice(): Extract a copy. Does not mutate the original.



```
arr.slice(1, 3)
```

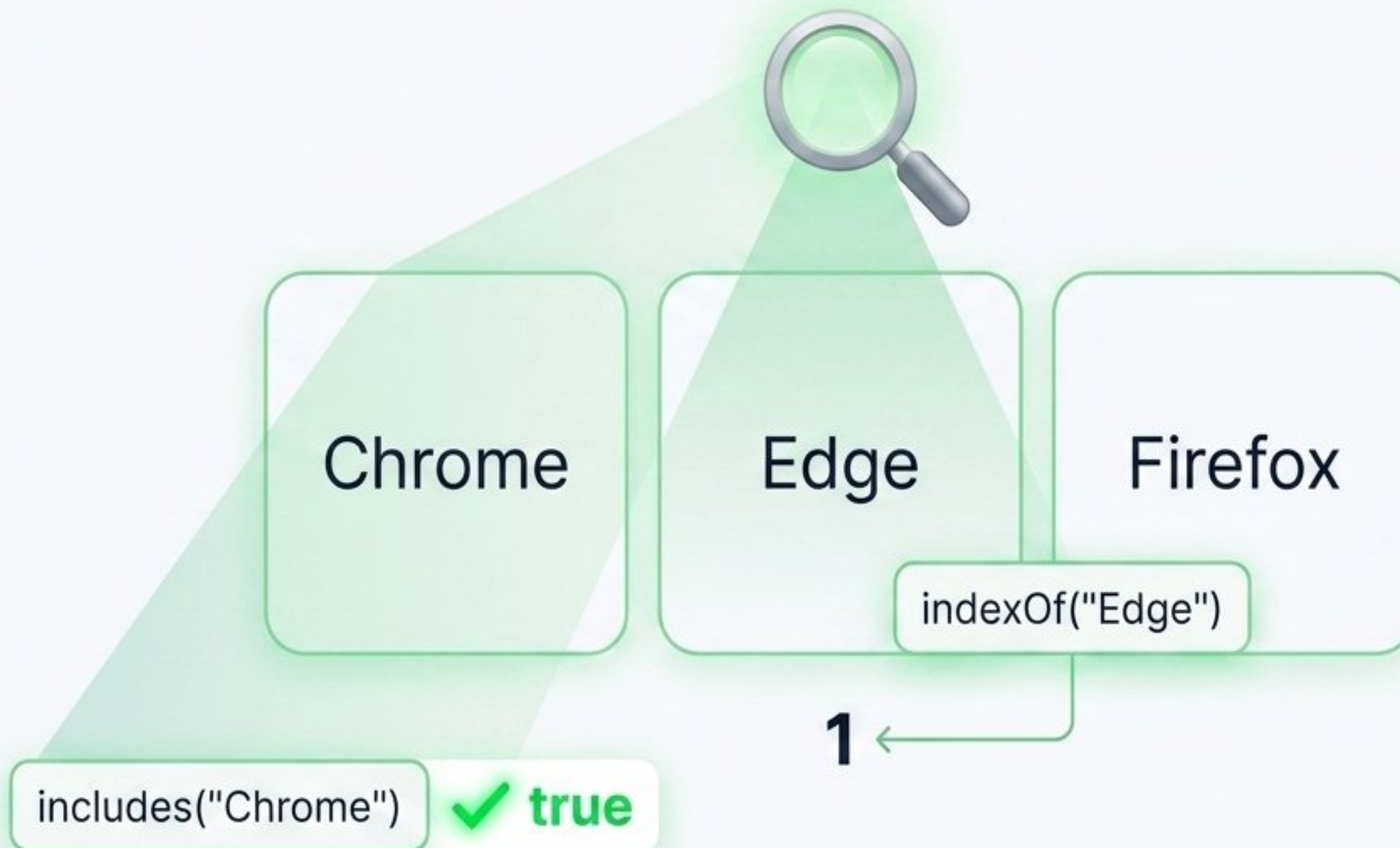
splice(): Add or remove elements. Mutates the original array.



```
arr.splice(2, 1)
```

Step-by-Step Search

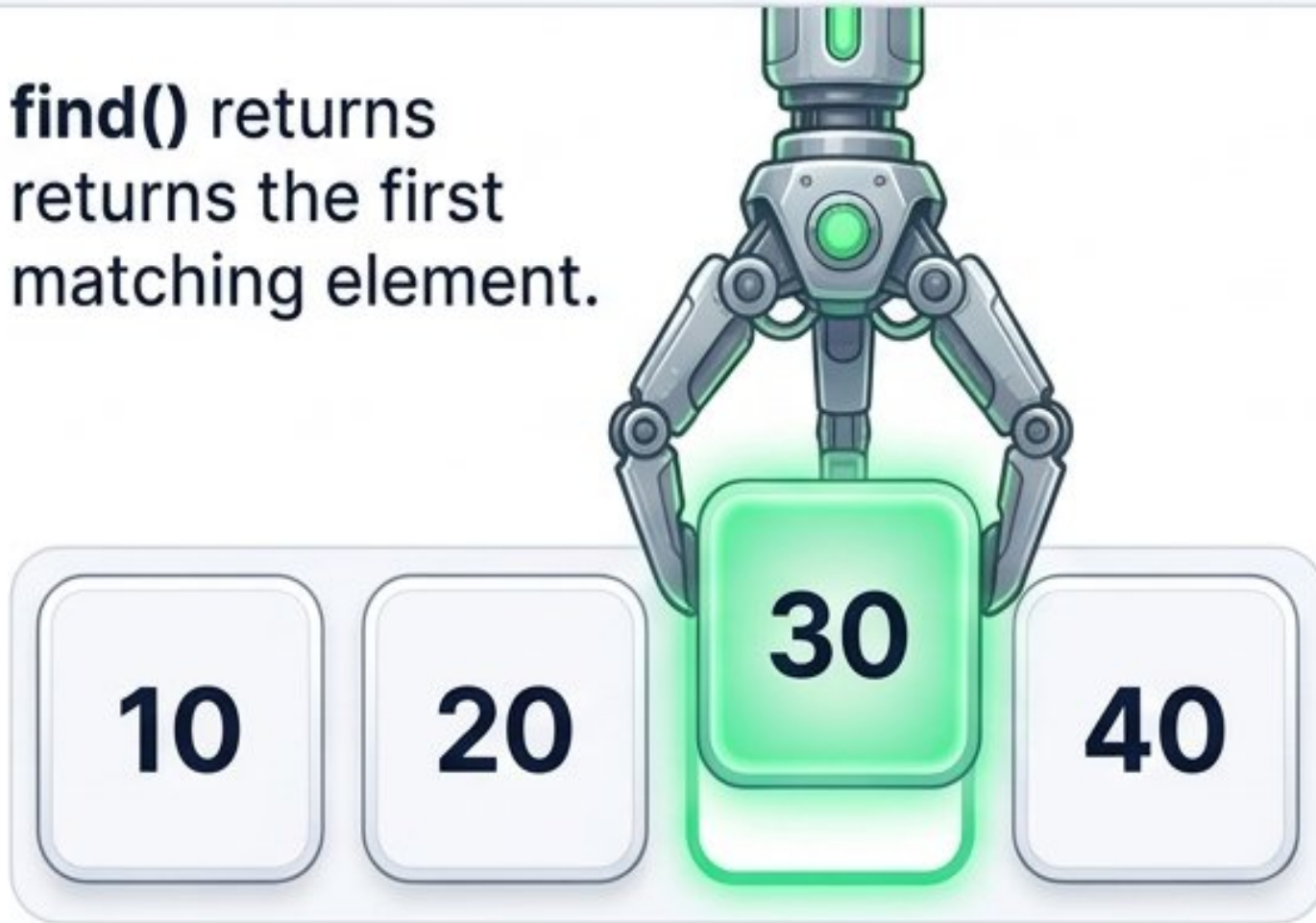
Instantly locate values or verify their existence within your dataset.



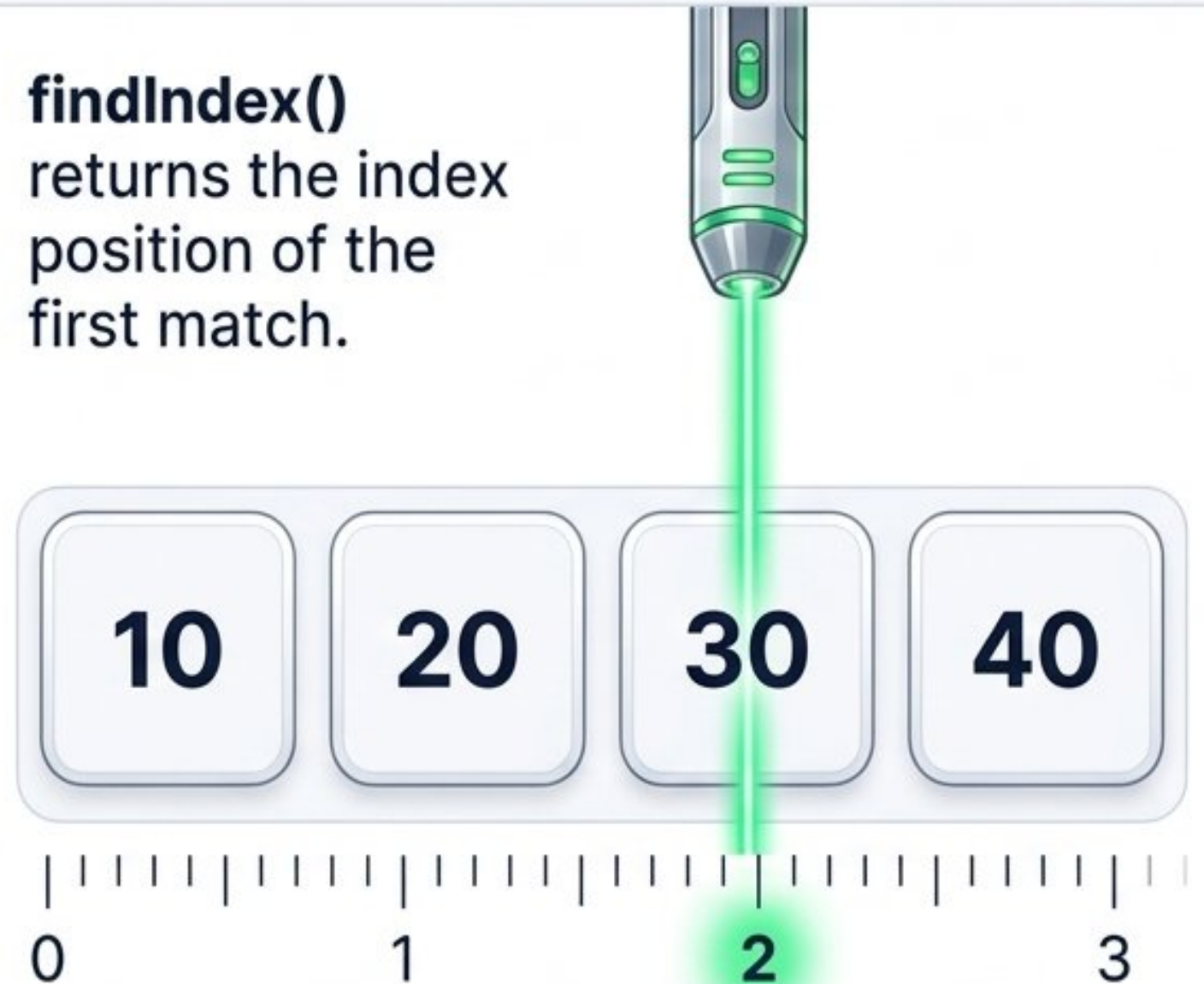
The Discovery Matrix: Extracting Data vs. Location

(num > 20)

find() returns
returns the first
matching element.

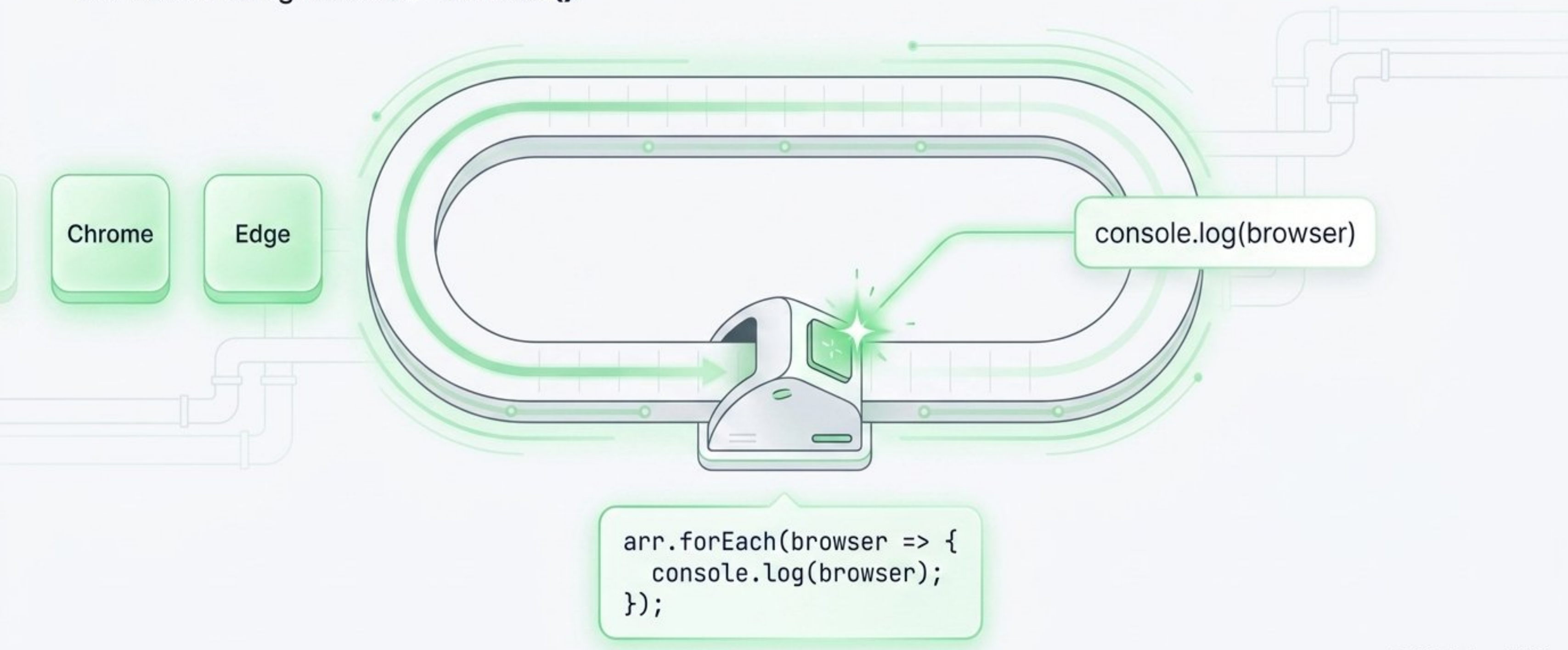


findIndex()
returns the index
position of the
first match.



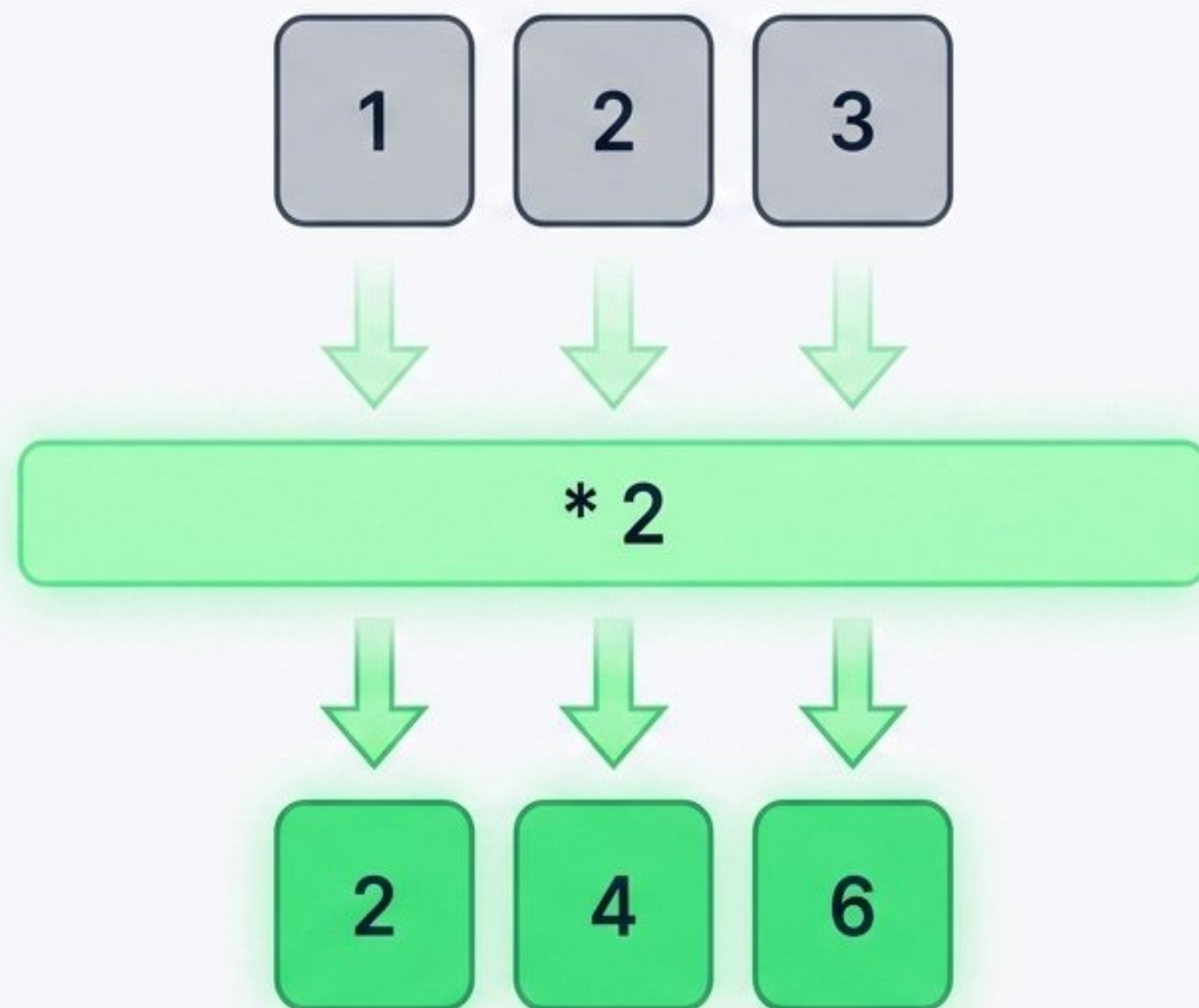
The Iteration Engine

Automate repetitive actions for every item in your collection using **for...of** or **forEach()**.



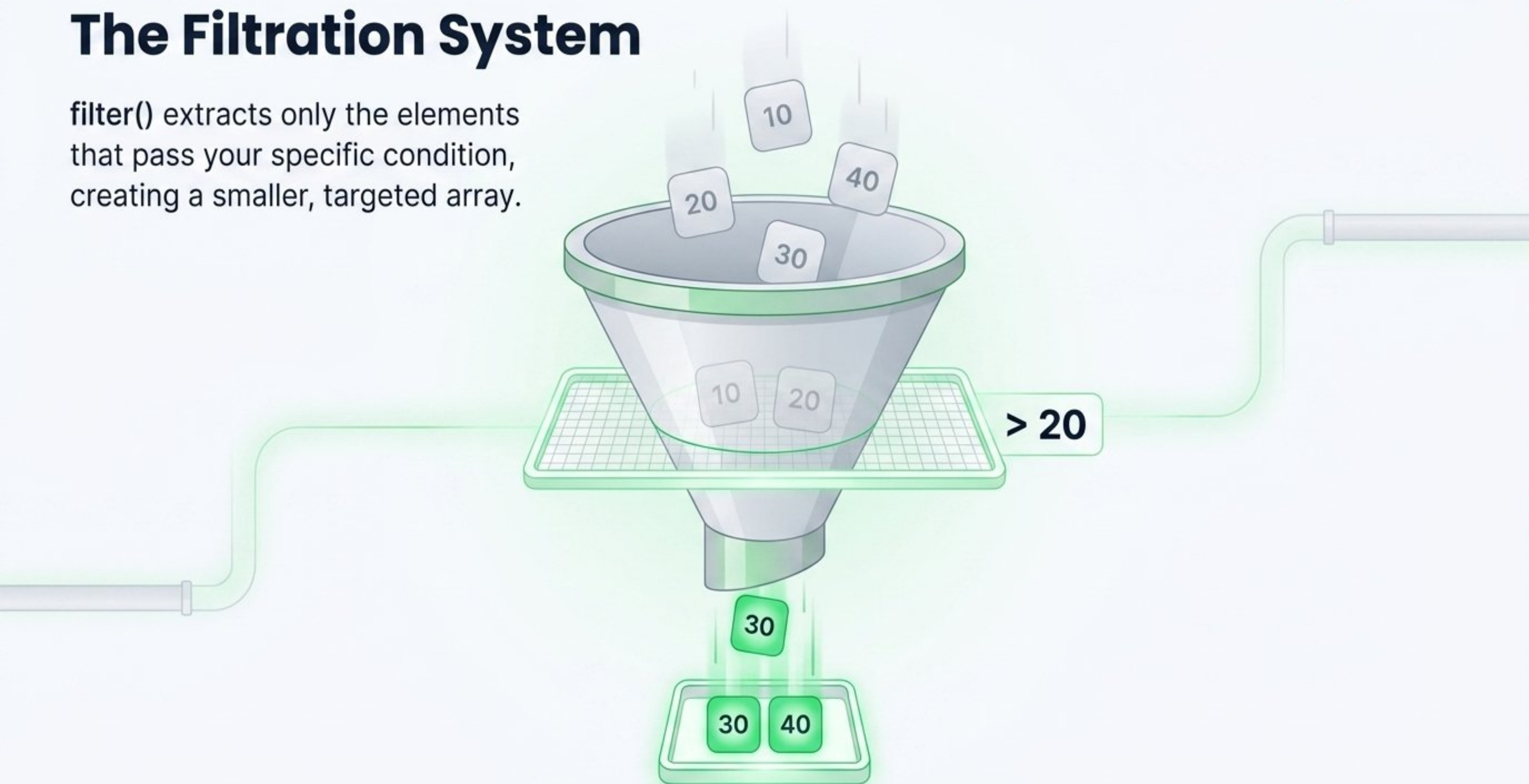
The Transformation Pipeline

map() transforms every element, outputting a completely new array of the exact same length.



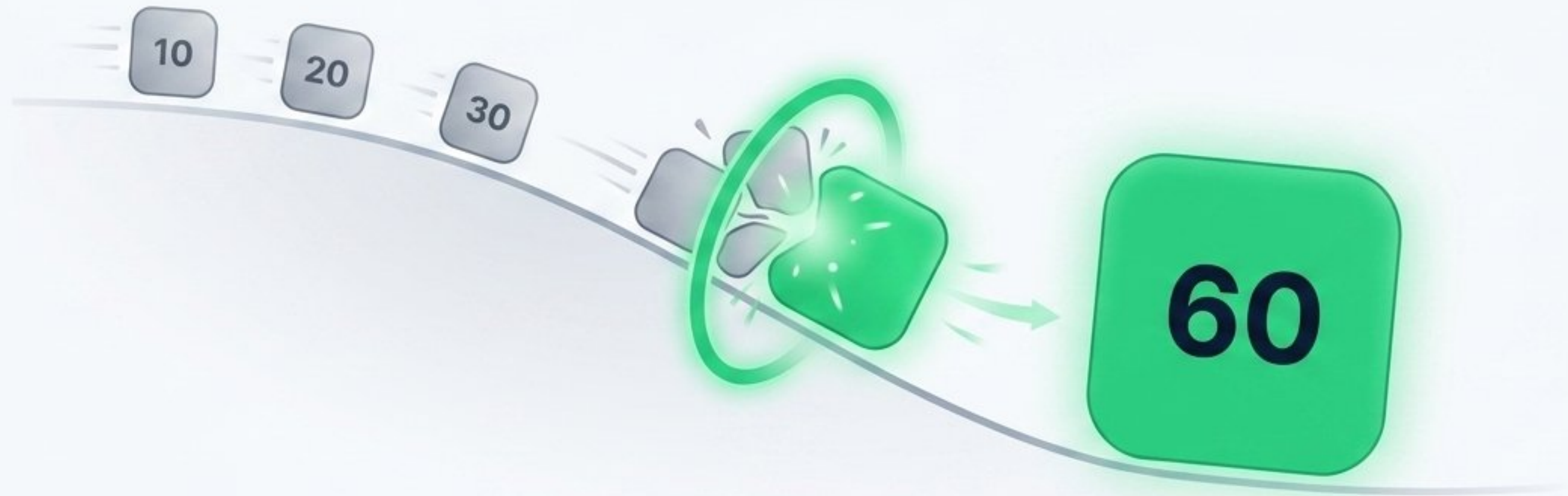
The Filtration System

`filter()` extracts only the elements that pass your specific condition, creating a smaller, targeted array.



The Accumulator

`reduce()` accumulates an entire array down to a single powerful value (like a sum or a total).



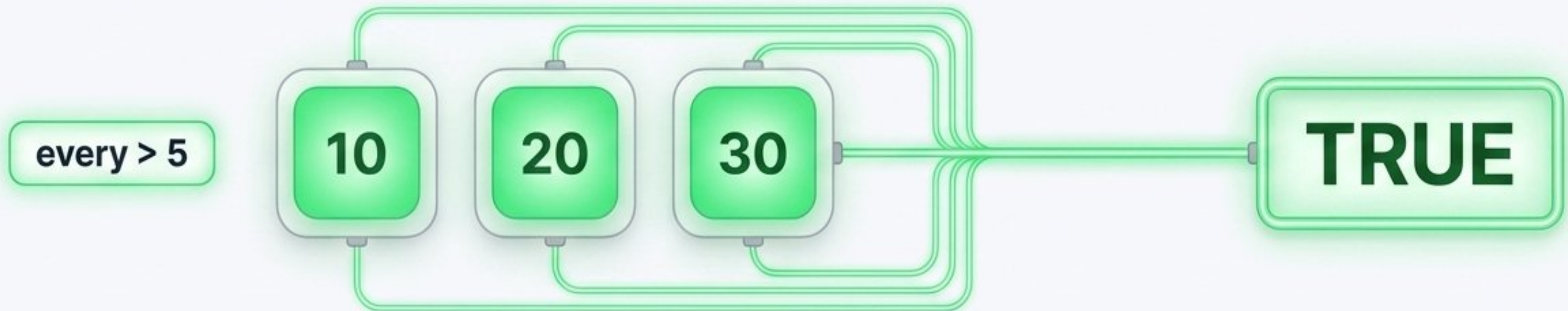
```
arr.reduce((sum, num) => sum + num, 0);
```

Logic Checks: Consensus vs. Any

some() - Requires at least one match to return true.



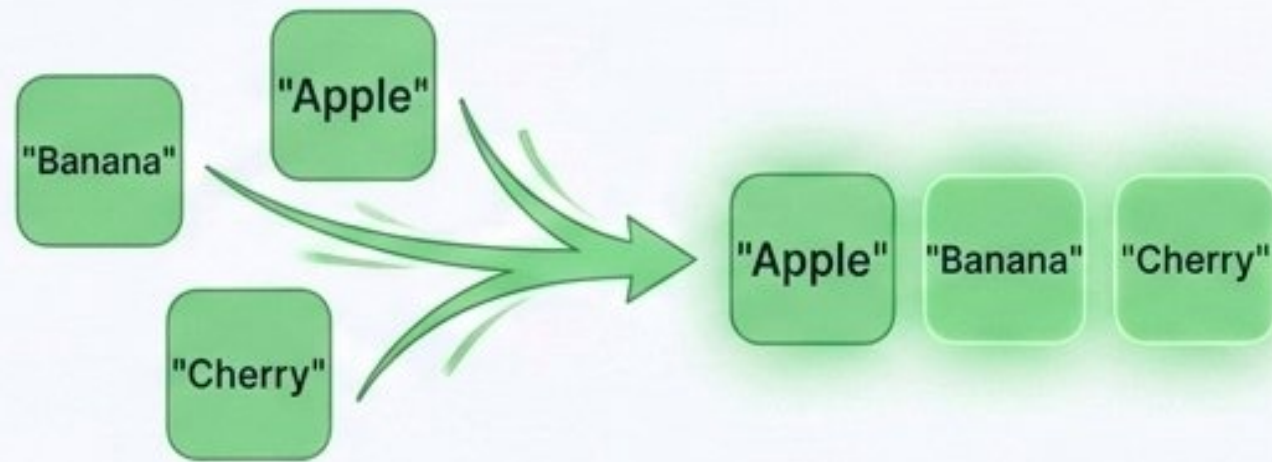
every() - Requires unanimous consensus from all elements to return true.



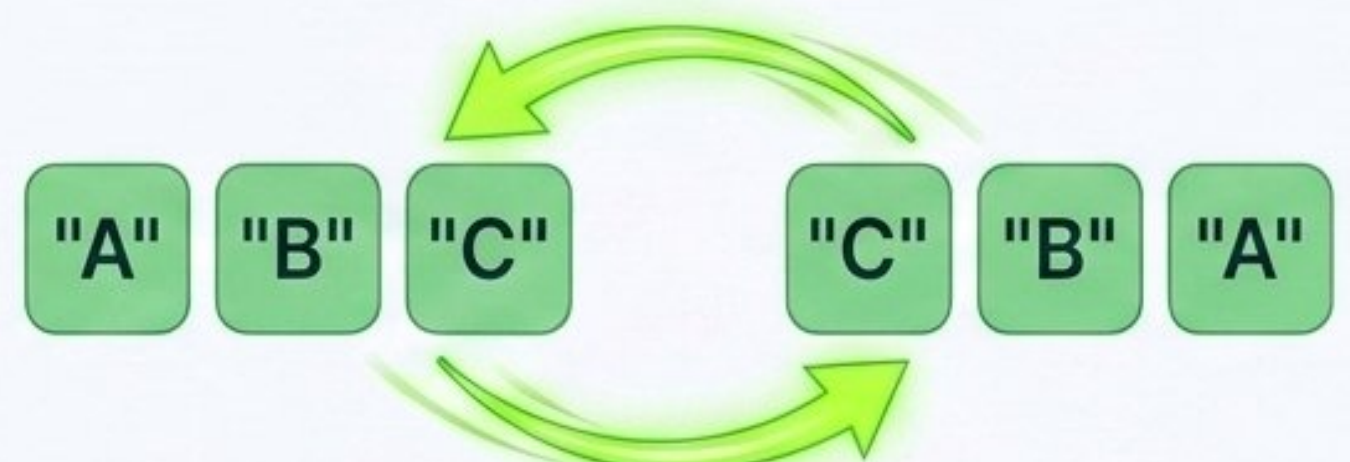
Utility & Organization

Essential methods to organize, merge, and format your array data.

sort()



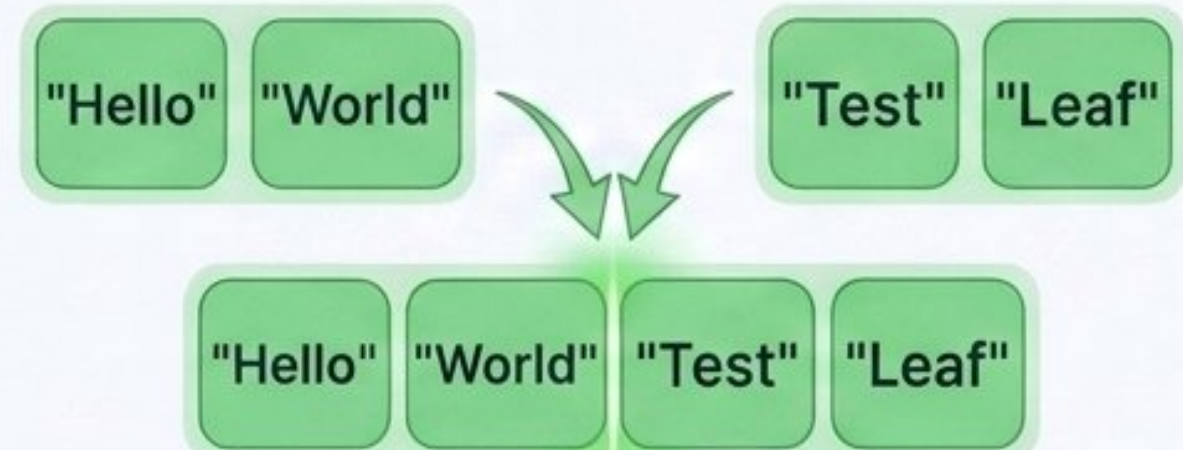
reverse()



join()



concat()



Real-World Use Cases: The Playwright Radar

Framework Development Focus. Master these 9 core functions—they are the most heavily utilized in modern test automation and API response handling.



The Array Arsenal



Your complete JavaScript array toolkit, categorized for instant recall.

